Challenge 1: Scurvy

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SCURVY

This is a weekly challenge from tidy Tuesday (Week 30)

https://github.com/rfordatascience/tidytuesday/blob/main/data/2023/2023-07-25/readme.md

Overview

James Lind's 1757 experiment is the first documented controlled clinical trial in history. Twelve sailors with scurvy were split into six treatment pairs (citrus, cider, vitriol, vinegar, seawater,a spice paste) and their symptoms scored after six days. Th challenge will attempted in both R and python

R Analysis

Load packages

```
library(tidyverse)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✔ dplyr     1.2.1     ✔ readr     2.2.0
✔ forcats   1.0.1     ✔ stringr   1.6.0
✔ ggplot2   4.0.3     ✔ tibble    3.3.1
✔ lubridate 1.9.5     ✔ tidyr     1.3.2
✔ purrr     1.2.1
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to
become errors
```

```
library(medicaldata)
library(tidytuesdayR)
library(janitor)
```

```

Attaching package: 'janitor'

The following objects are masked from 'package:stats':

    chisq.test, fisher.test
```

```
library(gtsummary)
```

Load data frame

The scurvy data frame is loaded from the tidytuesdayR Package

```
library(medicaldata)
scurvy <- medicaldata::scurvy

View(scurvy)
```

Inspection and cleaning

The dimensions of the data frame

```
dim(scurvy) #show both rows and columns
```

```
[1] 12  8
```

```
nrow(scurvy) # only rows
```

```
[1] 12
```

```
ncol(scurvy) # only columns
```

```
[1] 8
```

The 12 rows represent the 12 sailors that underwent the trial

The 8 columns defined as follows;

- study_id ; Participant ID
- treatment ; The treatments groups- cider, dilute_sulfuric_acid, vinegar, sea_water, citrus, purgative_mixture
- dosing_regimen_for_scurvy ; Dosing Regimen-1 quart per day; 25 drops of elixir of vitriol, three times a day; two spoonfuls, three times daily; half pint daily; two lemons and an orange daily; a nutmeg-sized paste of garlic, mustard seed, horseradish, balsam of Peru, and gum myrrh three times a day
- gum_rot_d6; Likert scale for severity of gum rot on day 6 (0_none, 1_mild, 2_moderate, 3_severe)
- skin_sores_d6 ; Likert scale for severity of skin sores on day 6 (0_none, 1_mild, 2_moderate, 3_severe)
- weakness_of_the_knees_d6 ; Likert scale for severity weakness of the knees on day 6 (0_none, 1_mild, 2_moderate, 3_severe)
- lassitude_d6 ; Likert scale for severity of Lassitude on day 6 (0_none, 1_mild, 2_moderate, 3_severe)
- fit_for_duty_d6 ; Whether the sailor is fit for duty (0_no, 1_yes)

Inspection

```
head(scurvy, n = 7)  # top 7 rows
```

```
# A tibble: 7 × 8
  study_id treatment           dosing_regimen_for_sc…¹ gum_rot_d6 skin_sores_d6
  <chr>    <fct>               <chr>                   <fct>      <fct>
1 001      cider               1 quart per day         2_moderate 2_moderate
2 002      cider               1 quart per day         2_moderate 1_mild
3 003      dilute_sulfuric_acid 25 drops of elixir of … 1_mild     3_severe
4 004      dilute_sulfuric_acid 25 drops of elixir of … 2_moderate 3_severe
5 005      vinegar             two spoonfuls, three t… 3_severe   3_severe
6 006      vinegar             two spoonfuls, three t… 3_severe   3_severe
7 007      sea_water           half pint daily         3_severe   3_severe
# i abbreviated name: ¹dosing_regimen_for_scurvy
# i 3 more variables: weakness_of_the_knees_d6 <fct>, lassitude_d6 <fct>,
#   fit_for_duty_d6 <fct>
```

```
tail(scurvy, n = 6) # bottom 6 rows
```

```
# A tibble: 6 × 8
  study_id treatment         dosing_regimen_for_scurvy  gum_rot_d6 skin_sores_d6
  <chr>    <fct>             <chr>                      <fct>      <fct>
1 007      sea_water         half pint daily            3_severe   3_severe
2 008      sea_water         half pint daily            3_severe   3_severe
3 009      citrus            two lemons and an orange … 1_mild     1_mild
4 010      citrus            two lemons and an orange … 0_none     0_none
5 011      purgative_mixture a nutmeg-sized paste of g… 3_severe   3_severe
6 012      purgative_mixture a nutmeg-sized paste of g… 3_severe   3_severe
# i 3 more variables: weakness_of_the_knees_d6 <fct>, lassitude_d6 <fct>,
#   fit_for_duty_d6 <fct>
```

Check structure

```
str(scurvy)
```

```
tibble [12 × 8] (S3: tbl_df/tbl/data.frame)
 $ study_id                : chr [1:12] "001" "002" "003" "004" ...
 $ treatment               : Factor w/ 6 levels "cider","citrus",..: 1 1 3 3 6 6 5 5 2
2 ...
 $ dosing_regimen_for_scurvy: chr [1:12] "1 quart per day" "1 quart per day" "25 drops
of elixir of vitriol, three times a day" "25 drops of elixir of vitriol, three times a
day" ...
 $ gum_rot_d6              : Factor w/ 4 levels "0_none","1_mild",..: 3 3 2 3 4 4 4 4 2
1 ...
 $ skin_sores_d6           : Factor w/ 4 levels "0_none","1_mild",..: 3 2 4 4 4 4 4 4 2
1 ...
 $ weakness_of_the_knees_d6 : Factor w/ 4 levels "0_none","1_mild",..: 3 3 4 4 4 4 4 4 1
1 ...
 $ lassitude_d6            : Factor w/ 4 levels "0_none","1_mild",..: 3 4 4 4 4 4 4 4 2
1 ...
 $ fit_for_duty_d6         : Factor w/ 2 levels "0_no","1_yes": 1 1 1 1 1 1 1 1 1 2 ...
```

No need to fix variable type as they are already fixed

Check for missing data

Overall

```
anyNA(scurvy)
```

```
[1] FALSE
```

```
colSums(is.na(scurvy))  # counts of missing values per ro
```

```
                study_id                treatment dosing_regimen_for_scurvy
                       0                        0                         0
            gum_rot_d6            skin_sores_d6  weakness_of_the_knees_d6
                       0                        0                         0
           lassitude_d6          fit_for_duty_d6
                       0                        0
```

In a particular column

```
anyNA(scurvy$lassitude_d6)  #
```

```
[1] FALSE
```

```
sum(is.na(scurvy$treatment))
```

```
[1] 0
```

No missing data noted anywhere in the dataframe

Check for duplicates

```
scurvy[duplicated(scurvy), ] # to see the duplicates
```

```
# A tibble: 0 × 8
# ℹ 8 variables: study_id <chr>, treatment <fct>,
#   dosing_regimen_for_scurvy <chr>, gum_rot_d6 <fct>, skin_sores_d6 <fct>,
#   weakness_of_the_knees_d6 <fct>, lassitude_d6 <fct>, fit_for_duty_d6 <fct>
```

```
sum(duplicated(scurvy))      # counts of the duplicates
```

```
[1] 0
```

No duplicated rows noted

Exploration

```
scurvy |> tabyl(treatment, fit_for_duty_d6)
```

```
            treatment 0_no 1_yes
                cider    2     0
               citrus    1     1
 dilute_sulfuric_acid    2     0
    purgative_mixture    2     0
              sea_water    2     0
                vinegar    2     0
```

```
# put in a better table
scurvy |>
  tbl_cross(
    row = treatment,
    col = fit_for_duty_d6,
    label = list(treatment ~ "Treatment",
                 fit_for_duty_d6 ~ "Fit for duty by day 6"
                 )
  )
```

	Fit for duty by day 6		
	0_no	1_yes	Total
Treatment			
cider	2	0	2
citrus	1	1	2
dilute_sulfuric_acid	2	0	2
purgative_mixture	2	0	2
sea_water	2	0	2
vinegar	2	0	2
Total	11	1	12

The sailors have been grouped according to whether they are fit for duty or not

After day 6; only one sailor is fit for duty

Next we shall group the sailors by severity of the symptoms

```
scurvy |>
  select(treatment, gum_rot_d6, skin_sores_d6, weakness_of_the_knees_d6, lassitude_d6)
         |>
  tbl_summary(
    by = treatment,
    label = list(
      gum_rot_d6 ~ "Gum rot",
      skin_sores_d6 ~ "Skin sores",
      weakness_of_the_knees_d6 ~ "Knee weakness",
      lassitude_d6 ~ "Lassitude"
    )
  )
```

Characteristic	cider N = 2[1]	citrus N = 2[1]	dilute_sulfuric_acid N = 2[1]	purgative_mixture N = 2[1]	sea_water N = 2[1]	vine N =
Gum rot						
0_none	0 (0%)	1 (50%)	0 (0%)	0 (0%)	0 (0%)	0 (0
1_mild	0 (0%)	1 (50%)	1 (50%)	0 (0%)	0 (0%)	0 (0
2_moderate	2 (100%)	0 (0%)	1 (50%)	0 (0%)	0 (0%)	0 (0
3_severe	0 (0%)	0 (0%)	0 (0%)	2 (100%)	2 (100%)	2 (100
Skin sores						
0_none	0 (0%)	1 (50%)	0 (0%)	0 (0%)	0 (0%)	0 (0
1_mild	1 (50%)	1 (50%)	0 (0%)	0 (0%)	0 (0%)	0 (0
2_moderate	1 (50%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0
3_severe	0 (0%)	0 (0%)	2 (100%)	2 (100%)	2 (100%)	2 (100
Knee weakness						
0_none	0 (0%)	2 (100%)	0 (0%)	0 (0%)	0 (0%)	0 (0
1_mild	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0
2_moderate	2 (100%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0
3_severe	0 (0%)	0 (0%)	2 (100%)	2 (100%)	2 (100%)	2 (100
Lassitude						
0_none	0 (0%)	1 (50%)	0 (0%)	0 (0%)	0 (0%)	0 (0

[1] n (%)

Characteristic	cider N = 2[1]	citrus N = 2[1]	dilute_sulfuric_acid N = 2[1]	purgative_mixture N = 2[1]	sea_water N = 2[1]	vine N =
1_mild	0 (0%)	1 (50%)	0 (0%)	0 (0%)	0 (0%)	0 (0
2_moderate	1 (50%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0
3_severe	1 (50%)	0 (0%)	2 (100%)	2 (100%)	2 (100%)	2 (100

[1] n (%)

Python Analysis

Interpretation